

Fungicide Resistance Management

Note: Not all indicated timings may be necessary for disease control; whereas additional applications may be necessary under favorable conditions for disease (see Treatment Timing Table). Suggested fungicide, bactericide, biological, and natural product groups are listed for each timing based on host phenology, weather monitoring, inoculum models, or environmental-disease forecasting models.

How to use this table:

1. Identify the disease(s) that need(s) to be managed. Know the disease history of the orchard, especially from the previous season.
2. Select one of the suggested fungicide groups. **Numbers separated by slashes are pre-mixtures, whereas numbers grouped by pluses are tank mixtures.** If several diseases need to be managed, select a group that is effective against all diseases. Refer to the [General Properties of Fungicides Used in Apples\(/agriculture/apple/General-Properties-of-Fungicides-Used-in-Apples/\)](#) for fungicides belonging to each FRAC group. Group numbers are listed in numerical order within the suggested disease management program.
3. Rotate groups for each application within a season and, if possible, use each group only once per season, except for multi-site mode-of-action materials (e.g., M 02) or natural products/biological controls (NP/BC).

Disease	Fall	Delayed dormant	Bloom			Spring
			Green tip/White bud	Pink bud/Full bloom	Petal Fall (PF)	Cover sprays
Fire blight	M 01	M 01	M 01, 24, 25, 41, P 01 ³	M 01, M 02 ^{2,4} , 24, 25, 41, BM 01, BM 02, BM 03, P 01, P 05, P 06	M 01 ¹ , 24, 25, 41, BM 01, BM 02, BM 03, P 01, P 05, P 06	M 01 ¹ , 24, 25, 41, BM 01, BM 02, BM 03, P 01, P 05, P 06, PGR ⁵
Scab	—	M 02	M 01 ¹ , M 02, M 03, 1, 3, 7, 9, 3/9, 7/11, 19, 52, U12, U12+M 01–03, 3/BM 01, BM 01, BM 02, BM 03, P 01, P 05, P 06	M 01 ¹ , M 02, M 03, 1, 3, 7, 9, 3/9, 7/11, 19, 52, U12, U12+M 01–03, 3/BM 01, BM 01, BM 02, BM 03, P 01, P 05, P 06	M 01 ¹ , M 02, M 03, 1, 3, 7, 9, 3/9, 7/11, 19, 52, U12, 3/BM 01, BM 01, BM 02, BM 03, P 01, P 05, P 06	M 01 ¹ , M 02, M 03, 1, 3, 7, 9, 3/9, 7/11, 19, 52, 3/BM 01, BM 01, BM 02, BM 03, P 01, P 05, P 06
Powdery mildew			M 02, 1, 3, 7, 9, 3/9, 7/11, 19, 52, 3/BM 01, BM 01, BM 02, BM 03, P 01, P 05, P 06	M 02, 1, 3, 7, 9, 3/9, 7/11, 19, 52, 3/BM 01, BM 01, BM 02, BM 03, P 01, P 05, P 06	M 02, 1, 3, 7, 9, 3/9, 7/11, 19, 52, 3/BM 01, BM 01, BM 02, BM 03, P 01, P 05, P 06	M 02, 1, 3, 7, 9, 3/9, 7/11, 19, 52, 3/BM 01, BM 01, BM 02, BM 03, P 01, P 05, P 06

**	Not registered, label withdrawn or inactive in California.
¹	Fixed copper (M 01a) bactericides (e.g., Kocide, Badge, Nordox, and ChamplON++) may cause phytotoxicity (russetting) when applied after full bloom. Other copper products (M 01b) with lower metallic copper equivalent (i.e., MCE) such as copper complexes (e.g., Cueva, Copper Count-N, etc.) and copper sulfate pentahydrate (e.g., CS-2005-only apples in CA, Phyton 27AG ^{**} , Instill, MasterCop, etc.) have been reported to be less phytotoxic with applications following bloom because of lower MCE (see specific registrant label concerning product rates and number of times each material can be applied during the growing season).
²	M 02 fungicides (e.g., liquid lime sulfur) that are registered for scab control have been used to thin flowers in Washington state (Section 24c label) with one to two applications between 20 and 80% full bloom and subsequently have reduced the total number of flowers and potential infections sites for fire blight (indirect effects on disease). No label is available in CA for this usage.
³	Acibenzolar-S-methyl (FRAC P 01) is a host plant defense inducer known to stimulate the salicylic acid pathway.

4	In general, sulfur compounds are fungicidal and may affect applications of fungal biocontrols (e.g., Blossom Protect); whereas copper may affect applications of bacterial biocontrols (e.g., Actinovate, Bloomtime Biological, Blight Ban, Double Nickel 55, and Serenade). Rotations must consider these factors.
5	Plant growth regulators (PGR) such as prohexadione calcium (e.g., Apogee) reduce shoot growth and thus, indirectly reduce the number of infections sites for fire blight (indirect effects on disease).

Acknowledgment: Adaskaveg et al., 2025. [Fungicides, Bactericides, Biocontrols, and Natural Products for Deciduous Tree Fruit and Nut, Citrus, Strawberry, and Vine Crops in California\(https://ipm.ucanr.edu/legacy_assets/PDF/PMG/fungicideefficacytiming.pdf\)](https://ipm.ucanr.edu/legacy_assets/PDF/PMG/fungicideefficacytiming.pdf).



UC IPM Pest Management Guidelines: Apple
UC ANR Publication 3432

[Authors and Credits \(./Authors-and-Credits\)](#)

Text Updated: 07/24